YERISWAMY DV

Product Sustaining & Quality Assurance Engineer | Orthopaedic Medical Devices

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Mobility & Work Authorization: Indian citizen | Open to roles across India | Open to employer-sponsored relocation across Europe, the UK, and Australia & New Zealand (ANZ).

PROFESSIONAL SUMMARY

Results-driven **Product Sustaining Engineer and Quality Assurance Lead** with **6+ years of specialized MedTech experience** delivering patient-specific orthopaedic implants (knee, hip, ankle), surgical resection guides, and high-precision tooling. Proven expertise establishing design control compliance under **FDA 21 CFR Part 820, ISO 13485:2016, EU MDR, TGA & Medsafe, and CDSCO/Medical Device Rules**. Directs cross-functional sustaining operations across a **16-member engineering team**, orchestrating Engineering Change Orders (ECN/ECO/DCR), root-cause investigations (CAPA/NCR via DMAIC, 5-Why, Fishbone), Test Method Validations (TMV), and Design History File (DHF) technical documentation. Closed **571 JIRA sustaining tickets** and **4 critical DCRs** with zero post-release escapes.

CORE COMPETENCIES & REGULATORY FRAMEWORKS

Regulatory Standards: EU MDR (2017/745 Annex II), UK MDR, ISO 13485:2016, FDA 21 CFR Part 820 (QMSR), ISO 14971 (Risk Management), TGA Essential Principles, CDSCO MDR, CE Technical Files, MDSAP.

Sustaining & Change Governance: Design Controls (21 CFR 820.30), DHF & DMR Governance, ECN / ECO / DCR Change Management, Closed-Loop CAPA, NCRs, Deviations, VOE, SCN.

Verification, Validation & Risk: Design V&V Protocol Authoring, Test Method Validation (TMV), Sample Size Justification, DFMEA & PFMEA Execution, Hazard Analyses, Traceability Matrices.

CAD, Tooling & Automation: SolidWorks (Advanced Assemblies, API Automation), Siemens NX, Creo, Autodesk Inventor, AutoCAD, GD&T (ASME Y14.5 / ISO 1101), High-Precision Tooling, Jigs & Fixtures.

Manufacturing & Metrology: Additive Manufacturing (3D Gyroid Lattice / SLM), CNC Machining, Optical Profile Projector, Vernier Calipers, Micrometres, Air Plug Gauges, CT Bone Segmentation.

PROFESSIONAL EXPERIENCE

Conformis India LLP (Acquired by restor3d, USA) — Aug 2021 to Present

Global Medical Technology Enterprise specializing in Patient-Specific Arthroplasty & Additive Orthopaedic Implants

Product Sustaining Engineer & Team Lead (Feb 2024 – Present)

- **Cross-Functional Sustaining Authority:** Lead sustaining engineering coordinating CAD Design, Additive/Subtractive Manufacturing, Process Engineering, and Quality Assurance across commercial implant and high-precision surgical tooling platforms.
- **Defect & Improvement Ticket Resolution:** Triaged, resolved, and verified **571 JIRA tickets** (design/CAD bugs and continuous improvement suggestions), releasing fixes across **4 controlled DCR cycles** with ~2-week turnaround cadences and zero post-release escapes.
- **Design Controls & Traceability:** Lead the compilation, review, and approval of Design History File (DHF) and Device Master Record (DMR) deliverables. Enforce end-to-end bidirectional traceability: *User Needs → Design Inputs → Design Outputs → Verification → Validation → Risk Controls* compliant with **FDA 21 CFR 820.30, ISO 13485:2016, EU MDR, and TGA guidelines**.
- **Design Transfer & Tooling Prints:** Author and approve Design Transfer packages ensuring seamless transition from CAD design freeze to shop-floor CAM nesting and additive manufacturing. Update high-precision tooling prints applying rigorous **ASME Y14.5 GD&T**.
- **V&V and Test Method Quality:** Review and approve design V&V test strategies, protocols, and sample size rationales; support **Test Method Validation (TMV)** ensuring measurement repeatability and gauge R&R.
- **ISO 14971 Risk Management:** Lead hazard identification and risk review boards; author and update **DFMEAs** and **PFMEAs** under ISO 14971 to mitigate patient safety and manufacturing failure modes throughout the device lifecycle.
- **Closed-Loop CAPA & Quality Investigations:** Spearhead non-conformance investigations (NCRs) and root cause analyses utilizing 5-Why, Fishbone, and DMAIC. Owned **1 critical CAPA to verified closure** and executed **5+ systematic containment audits** across 60 days of historical lots with 100% containment integrity.
- **Regulatory Audits & Submissions:** Partner with Regulatory Affairs to provide design assurance data, engineering rationales, and technical summaries for international regulatory filings (EU MDR Technical Files, US 510(k), TGA, CDSCO). Support audits.
- **Team Leadership:** Direct, mentor, and allocate daily tasks to a **16-member engineering team** across the Initial Process/Segmentation Team (9) and Implant/PSI Tooling Team (7). Conferred the **restor3d Certificate of Recognition (2024)**.

Team Lead – Product Design & Quality (Apr 2022 – Jan 2024)

- **Team Management:** Directed a 12-member engineering team delivering CT-scan anatomical 3D bone reconstructions, custom knee/hip implant variants, and high-precision surgical cutting blocks (PSI).
- **Technical Gatekeeping:** Conducted comprehensive engineering reviews on 100% of CAD deliverables in SolidWorks, verifying conformity against approved customer design inputs, bone resection boundaries, and additive frame constraints.

- **Standardization:** Standardized parametric CAD feature modeling trees, metadata labeling, and drafting checklists across the team, achieving a significant drop in first-pass pre-release rework.
- **Cross-Functional Review Boards:** Represented engineering in weekly Quality Review Boards (QRB) with US stakeholders, synchronizing design throughput with clinical delivery deadlines and ISO 14971 risk mitigations.

CAD & Quality Engineer – Anatomical Segmentation & Design (Aug 2021 – Mar 2022)

- **3D Bone Segmentation:** Reconstructed high-fidelity 3D anatomical surface models (STL/Parasolid) from clinical DICOM CT datasets, establishing femoral/tibial mechanical axes and reference planes for custom joint arthroplasty.
- **Parametric Modeling:** Performed custom cutting jig modifications and CAD part modeling in SolidWorks under controlled SOPs, ISO 13485 standards, and FDA 21 CFR 820 design controls.
- **Feasibility Analysis:** Conducted design feasibility evaluations, assessing joint clearance, resection angle boundaries, and mating clearances between patient bone surfaces and high-precision surgical tooling.

Career Break – Family Caregiver — Aug 2019 to Aug 2021

Stepped away from professional engineering to manage full-time personal family health commitments. Maintained active domain research in CAD/CAM modeling systems, medical device design controls, and high-precision tooling; successfully re-entered corporate MedTech engineering in August 2021, earning 2 leadership promotions in 2 years.

L&T Technology Services (Deputed via nFolks Software Technologies) — Jul 2018 to Aug 2019

Mechanical CAD Design Engineer (Orthopaedic Implants & High-Precision Tooling)

- **High-Precision Surgical Tooling Design:** Engineered patient-specific femoral and tibial resection guides, positioning jigs, and custom manufacturing holding fixtures for global orthopaedic programs using SolidWorks, Siemens NX, and Creo.
- **Production Drawings & GD&T:** Prepared detailed 2D manufacturing drawings for high-precision tooling adhering to **ASME Y14.5 / ISO 1101 GD&T**, defining datums, micron-level surface finishes, and tolerance stack-up limits for CNC machining.
- **Design Feasibility & Testing:** Collaborated with cross-functional R&D teams to conduct prototype feasibility studies, analyzing physical fit, ergonomics, and tool accessibility against rapid-prototyped anatomical bone models.
- **DHF Compilation:** Contributed objective verification evidence, engineering change documentation, and test summaries directly into client **Design History Files (DHF)**.

Rane Madras Pvt. Ltd — Jul 2017 to Jul 2018

Tool Room & Quality Control Trainee Engineer (Technical Apprenticeship)

- **Precision Tool Room & Sensored Fixture Design:** Modeled holding fixtures, jigs, and machining nests in SolidWorks and AutoCAD. Developed and commissioned a pneumatic/hydraulic sensored fixture for CNC machining centres to clamp steering gearboxes, integrating proximity/seating sensors to guarantee error-free component location.
- **Machining & Assembly:** Performed precision fabrication, manual bench fitting, and assembly of high-precision tooling components utilizing surface grinding machines, vertical milling machines, radial drills, and precision lathes.
- **Quality Metrology & Line Inspection:** Conducted First Piece Approvals (Initial Sample Inspection), in-process line quality audits, and final lot acceptance on active production lines.
- **Optical Metrology & Precision Gauging:** Operated an optical **Profile Projector (Optical Comparator)** to verify thread profiles, gear tooth forms, radii, and pitch diameters. Executed tight-tolerance quality inspections using digital vernier calipers, micrometres, height gauges, dial bore gauges, air plug gauges, and slip gauge sets.

KEY PROJECTS & TECHNICAL AUTOMATION

Proprietary Automation (SolidWorks API Review Macro): Developed a custom SolidWorks API macro programmatically inspecting assemblies for interference fit, bounding-box envelopes, cutting slot tolerances, and feature suppression states; generates an external timestamped HTML audit report (PASS/FAIL), cutting inspection time significantly.

Commercial Platform Sustaining (Identity Porous Knee Implant System): Managed sustaining and design transfer for the cementless porous knee platform incorporating 3D additive gyroid lattice biological fixation structures. Governed CAD updates, tolerance stack-ups, and DHF updates; closed 571 tickets across 4 DCR releases with 100% first-pass yield.

EDUCATION & TECHNICAL QUALIFICATIONS

Diploma in Tool and Die Making (4-Year Intensive Technical Engineering Program) — Jul 2014 to Jun 2018

Government Tool Room and Training Centre (GT&TC), Hospet, Karnataka, India

Comprehensive 4-year technical engineering curriculum focusing on precision mechanical tooling design, press tools, plastic injection moulds, die casting dies, advanced CAD/CAM, CNC programming, engineering metallurgy, GD&T, and industrial quality metrology.

Professional Equivalence: Backed by 6+ years of specialized corporate medical device engineering under US FDA 21 CFR Part 820 and ISO 13485 frameworks, satisfying practical engineering requirements for Senior Design Engineer and Lead roles across Indian R&D centers, European enterprises, and the Engineers Australia / Engineering New Zealand CDR framework.

CERTIFICATIONS & CONTINUOUS LEARNING

- **Medical Device Design & Development Control (ISO 13485)** – Udemy (Aug 2026)
- **ISO 14971: Risk Management for Medical Devices** – Udemy (In Progress)
- **Mastering Medical Devices Regulatory Affairs** – Udemy (In Progress)

- **Quality Management Systems Fundamentals** – Great Learning (Jan 2026)
- **Project Management Professional Practices** – Simplilearn SkillUP (Jul 2025)
- **Total Productive Maintenance (TPM)** – Larsen & Toubro (Feb 2019)
- **Certificate of Recognition (Excellence in Service & Dedication)** – restor3d / Conformis (Oct 2024)

TECHNICAL SKILLS & TOOLSET

CAD & Automation: SolidWorks (Advanced Assemblies, Surfacing, Sheet Metal), Siemens NX, Creo Parametric, Autodesk Inventor, AutoCAD, Autodesk Revit, SolidWorks API / VBA Macro Automation.

Medical Devices & Quality: Class I, IIa/IIb & III Orthopaedic Devices, Patient-Specific Implants (PSI), Surgical Cutting Jigs, Additive Manufacturing (SLM, Porous Lattice), DICOM CT Segmentation, ISO 13485, EU MDR Annex II, FDA 21 CFR 820, ISO 14971, CAPA, NCR, TMV, DHF/DMR.

Tooling & Metrology: High-Precision Tooling, Jigs & Sensored CNC Fixtures, ASME Y14.5 GD&T, Profile Projector, Micrometres, Vernier Gauges, Bore Gauges, Air Plug Gauges.

Languages: English (Full Professional Working Proficiency), Kannada (Native), Telugu (Fluent), Hindi (Fluent).